ISTE Educator Certification Program

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Google for Education

Teaching and Learning with Google Gemini

Lesson Overview
Lesson Title: AI Journeys: Ethics in Action
Grade Level(s): 11th – 12th grade
Subject Area(s) (Math, Reading, Science, etc.): IB Diploma Program Digital Society It can also support Interdisciplinary (Ethics, Technology, Global Issues)
Estimated Duration (e.g., 45 minutes, 2 class periods): 8 class periods (40 min each). *Educators are encouraged to adjust timings based on student needs and depth of discussion.*
What value does Gemini bring to this lesson? Gemini gives students on-demand research help, guided explanations, and feedback while they explore real AI dilemmas. It lets them personalize inquiries, compare policies, get writing support, and iterate on creative advocacy products. It encourages active student agency, ensuring Gemini serves as a tool for student exploration and creation rather than solely for teacher preparation. Used thoughtfully, Gemini boosts engagement, deepens ethical reflection, and models responsible AI use rather than serving as a novelty tool.
ISTE Standard(s) Alignment
Academic Standards (state, Common Core, NGSS, etc.) • **Concept 2.7 Values and ethics** Students explore values such as fairness, transparency, and accountability through discussions of AI dilemmas and by creating either a policy pitch or a creative campaign. • **Content 3.6 Artificial intelligence** During the Deep Dive phase, students use Gemini to summarize AI policies and identify ethical strengths and weaknesses, directly linking their research to the Artificial Intelligence content strand. • **Content 3.1 Data and data analysis** Student policy pitches and campaigns examine how data is collected, stored, analyzed, and governed, explicitly addressing key concepts from this content area. • **Context 4.5 Human knowledge** Class discussions address how generative AI reshapes our understanding of knowledge and learning, connecting clearly to the Human Knowledge context. • **Context 4.7 Social** Student projects highlight social implications of AI, including privacy, bias, and equity,

grounding their inquiries in authentic community contexts.

- **IB Command Terms (Analyze, Evaluate, Recommend):** Students use “Analyze” when examining policies, “Evaluate” in discussions about stakeholder impacts, and “Recommend” in their final policy pitches or campaigns.
- **Connection to IB Learner Profile:** This lesson promotes students becoming principled, thinkers, and communicators as they explore ethical AI dilemmas, critically analyze AI policies, and clearly express responsible-use recommendations.

ISTE Standards for Students (select from a list or provide relevant codes)

- **1.2 Digital Citizen:** Students reflect on ethical AI use in school and everyday life. They examine real-world examples, discuss responsible practices, and consider how their decisions impact others in digital environments.
- **1.3 Knowledge Constructor:** Students use Gemini to explore AI policies and ethical issues, evaluate sources, organize information, and construct a deeper understanding of responsible AI use.
- **1.4 Innovative Designer:** Students design solutions to authentic AI-related challenges. They create either a concise policy pitch or a creative campaign advocating for responsible AI practices.
- **1.6 Creative Communicator:** Students select effective formats (videos, infographics, audio clips, or slides) to clearly communicate their message. They share their work with an authentic school audience, gather peer feedback, and refine their products.

Learning Objectives

Learning Objectives:
Learning objectives are clear, measurable statements that describe what students should know, understand, or be able to demonstrate by the end of a learning experience. They use specific action verbs (following frameworks like Bloom's Taxonomy) to indicate the desired level of learning and provide a concrete way to assess student achievement, typically starting with "Students will be able to..." (SWBAT) followed by an observable action or outcome.

Students will be able to: Identify and explain key ethical principles related to AI in education and society.

Students will be able to: Analyze real AI dilemmas and evaluate their impacts on different stakeholders.

Students will be able to: Evaluate AI policies or guidelines using Gemini, identifying strengths and opportunities for improvement.

Students will be able to: Demonstrate responsible use of Gemini to gather, organize, and synthesize information for their chosen product.

Students will be able to: Design and publish either a concise policy pitch or a creative campaign advocating for responsible AI use, incorporating peer feedback.

Students will be able to: Reflect on how their understanding of AI ethics and generative tools has evolved, and outline personal guidelines for future AI use.

Use of Gemini

Use of Gemini:
Describe how students will directly use Gemini in this lesson to learn or produce something. The use of Gemini should be clear, purposeful, and designed to enhance student learning—not simply included as a novelty. Effective lessons will show how Gemini is embedded in creative and engaging activities that boost student interest, participation, and understanding. For example, focus on how students are using Gemini to:

- Deepen understanding with on-demand support, like explanations of challenging concepts and step-by-step guidance
- Prepare for exams by creating personalized practice quizzes based on hard-to-grasp concepts
- Brainstorm and research ideas, get feedback on writing, and generate citations
- Think critically about a topic

How will students use Gemini during the lesson?

Students use Gemini purposefully and meaningfully at each step, enhancing their learning, supporting creativity, and deepening reflection about AI ethics.

1. Onboarding and Exploring Gemini

Students learn essential Gemini features (voice input, image upload, double-check, citations). They practice writing effective prompts using the PARTS framework (Purpose, Audience, Role, Task, Steps), experiencing how clear prompts yield focused, informative responses.

2. Deepening Understanding of AI Policies

Students upload or link an AI policy into Gemini and prompt it to summarize key points clearly. They use Gemini to identify ethical strengths and opportunities for improvement, supporting their analysis with credible, cited evidence.

3. Creating and Refining Products

- Policy Pitch Teams: Use Gemini to outline their policy brief, generate supporting evidence, and refine persuasive language and citations.
- Creative Campaign Teams: Use Gemini for brainstorming campaign ideas, developing clear messages, and checking scripts for clarity and bias.
  In both cases, students iteratively prompt Gemini to enhance their work and ensure factual accuracy.

4. Reflecting on Learning and Ethical Thinking

Students ask Gemini to suggest reflection questions about how their understanding of ethical AI has evolved. Each student chooses one suggested question and answers it in a reflective

journal entry (text, audio, or video).

Across all activities, Gemini provides personalized, on-demand support, guiding students in research, writing, critical thinking, and reflection, rather than serving as a simple novelty or passive tool.

Procedure

Procedure:
Outline the step-by-step flow of the lesson, ensuring it is well-organized, easy to follow, and uses clear, concise language. Each part of the lesson should demonstrate a purposeful use of Gemini, with a focus on how students interact with the tool to deepen learning—not as a novelty. Be sure to include any materials needed and how learning will be assessed through both formative and/or summative methods.

Opening / Engagement Activity (40 min)
Students begin by exploring a short real-world AI dilemma in education, such as AI used for grading, behavior tracking, or content generation. They choose how to engage with the case:

- Option 1: Read a brief written version
- Option 2: Listen to an audio version

In small groups, students discuss:

- What is the ethical issue in this scenario?
- Who might be impacted and how?
- What questions does this raise about how AI is used in schools?

After the discussion, students individually complete a 3-2-1 Reflection:

- 3 things they learned or found important
- 2 questions they still have
- 1 connection to their own experience

This activity surfaces students' current understanding, builds curiosity, and sets the stage for deeper inquiry using Gemini in the next phase.

Detailed Step by Step Lesson Structure featuring student use of Gemini:

Onboarding Gemini (40 min)

The class begins by watching the video "5 Essential to Know about Generative AI from Google" to explore how students can use AI tools responsibly and creatively. After the video, the teacher introduces Gemini through a live demo or visual guide, highlighting key features such as voice input, image upload, citation links, double-check, and chat sharing.

Students are introduced to the PARTS prompt-writing framework (Persona, Act, Recipient, Theme, Structure) as a way to guide clearer, more purposeful communication with Gemini. The teacher and class work together to revise a vague sample prompt and compare the results in Gemini to see how clarity improves the output. It's important to emphasize that

prompt writing is an iterative process; students should expect to refine their prompts based on initial Gemini responses and peer feedback.

Next, students create their own prompt based on the case study discussed earlier. Example: "*You are a student preparing to join a class discussion about the Hingham High School case. Explain one ethical issue this situation raises about using AI in school. Write your response in a way that's easy to understand for other students, and include one example to help make your point clear.*"

Before running their prompt in Gemini, students share it with a partner or small group for quick feedback:

- Is the purpose of the prompt clear?
- Does it give enough context for Gemini to respond well?
- Will the answer help you think more deeply about the ethical issue?

Students then run their revised prompt, read the response, and save it along with the citation in their notes for future reference.

This stage gives students a foundation for using Gemini thoughtfully and prepares them to explore more complex ideas in the next phase.

Deep Dive/Research (80 min)

Students select an AI-related policy or guideline to explore, such as their school's acceptable use policy or the IB's AI guidance. Using a structured task sheet, they analyze the document through an ethical lens, identifying its purpose, supported values, possible gaps, and impact on different groups in a school setting.

Gemini is used to support understanding and critical thinking. Students may ask Gemini to:

- Explain complex or unclear sections of the policy
- Define key terms like fairness, bias, or accountability
- Identify how the policy supports or limits student rights or responsibilities
- Explore how the policy connects to the ethical issues raised in the Hingham High School case

Example prompt: *"Help me summarize this AI policy and identify its key ethical principles regarding fairness and privacy. Then, suggest one specific area where this policy could be improved to better address ethical AI use, providing a brief justification."*

Students add relevant insights from Gemini into their task sheet, but focus on analyzing the policy in their own words. They compare findings with peers in small groups to expand their understanding and begin identifying key messages they may want to communicate in the next phase.

Product Development (80 mins)

Students apply their research to design a product that communicates their message about responsible AI use in schools. They choose one of two options:

- A **policy pitch** that presents a clear, actionable recommendation for improving AI guidelines
- A **creative campaign** such as a video, infographic, podcast, or social media content (e.g. short-form video, carousel, or image post) that raises awareness about ethical AI use and its impact on students

Students use Gemini to support their creative process. Depending on their product, they may use it to:

- Brainstorm how to communicate their message clearly and persuasively
- Generate or revise outlines, scripts, captions, or taglines
- Ask for feedback on tone, clarity, or structure
- Check for bias, suggest stronger examples, or improve wording

Example Gemini prompt: "*You are a student creating a social media post to raise awareness about AI in schools. Suggest three ways to make my message clear and engaging for other students. Help me write a short caption and check if the message is fair and respectful.*"

Throughout this process, students document how Gemini supported their work. They use feedback from peers and the teacher to revise their product and strengthen clarity, accuracy, and ethical impact.

By the end of this stage, students are ready to share their message with an authentic audience and reflect on their learning journey.

Share (40 mins)

Students present their final products, either a policy pitch or a creative campaign, to the class. Projects are also uploaded to a shared digital space such as FigJam, where classmates leave peer feedback using the TAG protocol:

- Tell something you like
- Ask a question
- Give a suggestion

This step gives students the opportunity to communicate their ideas to an authentic audience, receive constructive input, and engage in thoughtful conversations about ethical AI use in schools.

Reflect (20-25 mins)

After reviewing peer feedback, students take time to reflect on their learning. They write a prompt in Gemini using the PARTS framework. For example: *"You are a high school student finishing a project on AI ethics. Suggest three reflection questions I can use to think about how my views on*

ethical AI use in schools have changed. Make the questions clear and easy to answer in a journal format."

Students select one of Gemini's suggested questions and respond with a short journal entry reflection. They may submit this as written text, audio, or video. Each student also adds one personal guideline they plan to follow when using AI in the future.

Closure / Wrap-Up (20-25 min)

To close the project, the class works together to create a Student Manifesto for Ethical AI Use in Schools. This shared artifact reflects what they believe, value, and want to see in how AI is used for learning.

The teacher facilitates a short brainstorm based on prompts such as:

- What matters most when using AI in education?
- What should never be forgotten?
- What do students need from schools to use AI responsibly and creatively?

Students co-write their manifesto in small groups or as a whole class. They may choose to present it as a written statement, visual poster, video, or spoken word piece.

The manifesto can be shared with other classes, posted in the school, or presented to staff, school leadership, or even relevant community stakeholders as a student contribution to shaping future AI policy and culture.

Assessment

Assessment Strategy:

Student learning is assessed in two ways:

- **Group Assessment (Summative):** The final product, either a policy pitch or a creative campaign, is assessed using a rubric aligned with IB DP Digital Society expectations. The rubric focuses on how students apply their understanding of ethical issues, analyze digital systems and their impacts, consider multiple stakeholder perspectives, and communicate ideas effectively.
- **Individual Reflection (Formative)**: Each student completes a personal reflection supported by Gemini. This is not formally graded but can be used to give feedback on Approaches to Learning (ATL) skills such as critical thinking, ethical thinking, and self-management.

Peer feedback using the TAG protocol and contributions to the class manifesto are part of collaborative engagement and classroom culture, but are not included in the formal assessment.

Digital Tools/Resources Used:

- Gemini (accessed via school-provided Google Workspace accounts with Gemini features enabled) to support ethical inquiry, policy analysis, product development, and personal reflection
- FigJam to publish student work and gather peer feedback
- Google Docs or Slides for organizing research, drafting content, and collaborative planning
- Video and audio tools such as Google Vids, Canva, or Adobe Express for creating final products or recording reflections
- Curated resources such as school AUPs, IB AI guidance, and sample policies for analysis

Evidence of Learning:

- Gemini prompts and responses generated by students
- Completed group product (policy pitch or creative campaign)
- Research and planning captured in the project planner
- Peer feedback provided through TAG
- Individual reflection and personal AI use guideline
- Class-created AI ethics manifesto

Differentiation and Support

Beginning Learner Modifications:

- Provide simplified or summarized versions of case study and policy documents to support comprehension
- Use visuals and graphic organizers to scaffold research, ethical analysis, and product planning
- Pair students with supportive classmates for collaborative thinking and language support
- Allow students to record ideas orally before writing, and provide translation or language support when needed
- Provide optional sentence stems or writing frames to support prompt creation, ethical reasoning, and reflection. These may be especially useful for students developing academic English or those who are

Advanced Learners Modifications:

- Encourage deeper analysis of complex or conflicting AI policies
- Invite students to explore global implications of ethical AI beyond education
- Support integration of IB concepts such as power, equity, and digital identity into their product or reflection
- Offer opportunities to lead small group discussions or facilitate peer feedback sessions
- Allow for multi-layered campaign creation, including stakeholder perspectives or policy simulations

new to IB-style analysis.	

Support Resources

- Student Template
- Assessment Rubric
- Case Study (Source)
- AI Policies
- Teach Responsible Use of AI
- Prompting Guide 101
- AI Literacy for Students
- 5 essentials to know about generative AI from Google

ISTE Certified Educator Information

Educator Name: Greta Sandler

School / Organization: St. Catherine's Moorlands School